WHY STUDY *Imperfections in Solids?*

The properties of some materials are profoundly influenced by the presence of imperfections. Consequently, it is important to have a knowledge about the types of imperfections that exist and the roles they play in affecting the behavior of materials. For example, the mechanical properties of pure metals experience significant alterations when the metals are alloyed (i.e., when impurity atoms are added)—for example, brass (70% copper–30% zinc) is much harder and stronger than pure copper (Section 7.9).

Also, integrated-circuit microelectronic devices found in our computers, calculators, and home appliances function because of highly controlled concentrations of specific impurities that are incorporated into small, localized regions of semiconducting materials (Sections 18.11 and 18.15).

Learning Objectives

After studying this chapter, you should be able to do the following:

1. Describe both vacancy and self-interstitial crystalline defects.
2. Calculate the equilibrium number of vacancies in a material at some specified temperature, given the relevant constants.
3. Name the two types of solid solutions and provide a brief written definition and/or schematic sketch of each.
4. Given the masses and atomic weights of two or more elements in a metal alloy, calculate the weight percent and atom percent for each element.
5. For each of edge, screw, and mixed dislocations:
   (a) describe and make a drawing of the dislocation,
   (b) note the location of the dislocation line, and
   (c) indicate the direction along which the dislocation line extends.
6. Describe the atomic structure within the vicinity of (a) a grain boundary and (b) a twin boundary.

4.1 INTRODUCTION

Thus far it has been tacitly assumed that perfect order exists throughout crystalline materials on an atomic scale. However, such an idealized solid does not exist; all contain large numbers of various defects or **imperfections.** As a matter of fact, many of the properties of materials are profoundly sensitive to deviations from crystalline perfection; the influence is not always adverse, and often specific characteristics are deliberately fashioned by the introduction of controlled amounts or numbers of particular defects, as detailed in succeeding chapters.

imperfection

A *crystalline defect* refers to a lattice irregularity having one or more of its dimensions on the order of an atomic diameter. Classification of crystalline imperfections is frequently made according to the geometry or dimensionality of the defect. Several different imperfections are discussed in this chapter, including **point defects** (those associated with one or two atomic positions); linear (or one-dimensional) defects; and interfacial defects, or boundaries, which are two-dimensional. Impurities in solids are also discussed, because impurity atoms may exist as point defects. Finally, techniques for the microscopic examination of defects and the structure of materials are briefly described.

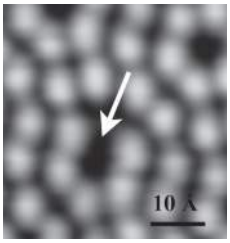
point defect

Point Defects

4.2 VACANCIES AND SELF-INTERSTITIALS

The simplest of the point defects is a **vacancy,** or vacant lattice site, one normally occupied but from which an atom is missing (Figure 4.1). All crystalline solids contain vacancies, and, in fact, it is not possible to create such a material that is free of these

vacancy



Scanning probe micrograph that shows a vacancy on a (111)-type surface plane for silicon. Approximately 7,000,000×. (Micrograph courtesy of D. Huang, Stanford University.)

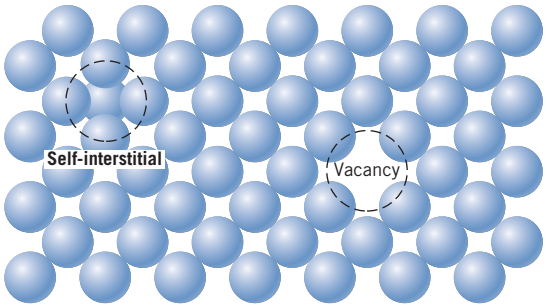


Figure 4.1 Two-dimensional representations of a vacancy and a self-interstitial. (Adapted from W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, John Wiley & Sons, 1964. Reproduced with permission of Janet M. Moffatt.)

defects. The necessity of the existence of vacancies is explained using principles of thermodynamics; in essence, the presence of vacancies increases the entropy (i.e., the randomness) of the crystal.

The equilibrium number of vacancies N_v for a given quantity of material (usually per meter cubed) depends on and increases with temperature according to

Temperature dependence of the equilibrium number of vacancies

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right) \tag{4.1}$$

In this expression, N is the total number of atomic sites (most commonly per cubic meter), Q_v is the energy required for the formation of a vacancy (J/mol or eV/atom), T is the absolute temperature in kelvins,[1] and k is the gas or **Boltzmann's constant.** The value of k is 1.38×10^{-23} J/atom·K, or 8.62×10^{-5} eV/atom·K, depending on the units of Q_v.[2] Thus, the number of vacancies increases exponentially with temperature—that is, as T in Equation 4.1 increases, so also does the term $\exp(-Q_v/kT)$. For most metals, the fraction of vacancies N_v/N just below the melting temperature is on the order of 10^{-4}—that is, one lattice site out of 10,000 will be empty. As ensuing discussions indicate, a number of other material parameters have an exponential dependence on temperature similar to that in Equation 4.1.

Boltzmann's constant

self-interstitial

A **self-interstitial** is an atom from the crystal that is crowded into an *interstitial site*—a small void space that under ordinary circumstances is not occupied. This kind of defect is also represented in Figure 4.1. In metals, a self-interstitial introduces relatively large distortions in the surrounding lattice because the atom is substantially larger than the interstitial position in which it is situated. Consequently, the formation of this defect is not highly probable, and it exists in very small concentrations that are significantly lower than for vacancies.

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Tutorial Video: Computation of the Equilibrium Number of Vacancies

[1]Absolute temperature in kelvins (K) is equal to °C + 273.

[2]Boltzmann's constant per mole of atoms becomes the gas constant R; in such a case, $R = 8.31$ J/mol·K.